

Fireflies in Union

How Simple Timing Rules Create a Collective Rhythm

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1 The Puzzle in the Forest

What do you think happens when hundreds of fireflies gather and begin flashing, each one following its own slightly different internal clock? At first, the scene may appear completely random. Yet, after observing and responding to nearby flashes, the individuals gradually adjust their timing until the entire group begins to pulse in a shared rhythm—with no leader, no conductor, and no central control [1]. That’s beautiful, right?

This is an example of synchronization: interacting systems that initially behave differently begin to follow a common rhythm. The same broad idea appears in pendulum clocks, cardiac cells, electrical generators, and communication networks.

The photograph in fig. 1 records light trails rather than a mathematical graph. The model below asks a simpler question:

What is the smallest timing rule that can turn many independent flash cycles into one coherent display?



Figure 1: Firefly light trails in a garden. Photograph by Bernd Thaller, via Wikimedia Commons.

1.1 A Firefly as a Moving Clock Hand

Represent firefly i by a phase

$$\theta_i(t) \in [0, 2\pi)$$

One complete trip around the phase circle is one flash cycle. The firefly flashes when its phase reaches 2π , after which the phase begins again at 0. If it is isolated, then

$$\frac{d\theta_i}{dt} = \omega_i, \quad \theta_i(t) = (\theta_i(0) + \omega_i t) \pmod{2\pi}, \quad (1)$$

where ω_i is its natural angular frequency. A larger ω_i means a shorter flash period

$$T_i = \frac{2\pi}{\omega_i} \quad (2)$$

Real fireflies emit brief pulses, so this continuous-clock picture is an approximation.¹

2 Listening and Adjusting

2.1 The Kuramoto Rule

A widely used model for interacting phases is

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i), \quad i = 1, \dots, N \quad (3)$$

The three parts of eq. 3 have direct meanings:

¹Pulse-coupled models represent each flash as a discrete event and are often closer to the biological mechanism[2]

Internal clock

ω_i is the rate the firefly would follow alone.

Coupling strength

$K \geq 0$ measures how strongly flashes influence one another

Timing correction

$\sin(\theta_j - \theta_i)$ tells firefly i whether another phase is ahead or behind

Table 1 explains the correction term. The table is simple, but it contains the central intuition of the model.

Table 1: How a neighboring firefly affects firefly i .

Relative pOsition	Phase Difference	Sine value	Effect on firefly i
Nrighbour ahead	$0 < \theta_j - \theta_i < \pi$	Positive	Speeds up its phase
Phases aligned	$\theta_j - \theta_i = 0$	zero	Makes no correction
Neighbour behind	$-\pi < \theta_j - \theta_i < 0$	Negative	Slows down its phase
Oposite phases	$\theta_j - \theta_i = 0$	Zero	Produces no preferred direction of correction.

The sine function is useful because it is smooth, periodic, and changes sign when the order of two phases is reversed.

3 Two Fireflies: An Exact Answer

The full model may contain hundreds of phases, but the case $N = 2$ can be solved directly. Define

$$\Delta = \theta_2 - \theta_1, \quad \Delta\omega = \omega_2 - \omega_1.$$

For two fireflies, eq. 3 becomes

$$\frac{d\omega_1}{dt} = \omega_1 + \frac{K}{2} \sin(\theta_2 - \theta_1) \quad (4)$$

$$\frac{d\omega_2}{dt} = \omega_2 + \frac{K}{2} \sin(\theta_1 - \theta_2) \quad (5)$$

Subtracting the first equation from the second gives

$$\frac{d\Delta}{dt} = \Delta\omega + \frac{K}{2} \sin(-\Delta) - \frac{K}{2} \sin(\Delta) \quad (6)$$

$$= \Delta\omega - K \sin \Delta \quad (7)$$

The two fireflies are phase locked when Δ becomes constant. Setting the left side of eq. 7 to zero

$$\sin \Delta^* = \frac{\Delta\omega}{K} \quad (8)$$

A real solution exists only when

$$|\Delta\omega| \leq K. \quad (9)$$

Thus, coupling can overcome a small mismatch between internal clocks, but not an arbitrarily large one.

4 From a Pair to a Population

4.1 Measuring Agreement

To measure how closely N phases are grouped, define the order parameter

$$r = \frac{1}{N} \sqrt{\left(\sum_{j=1}^N \cos \theta_j \right)^2 + \left(\sum_{j=1}^N \sin \theta_j \right)^2}, \quad 0 \leq r \leq 1 \quad (10)$$

Each phase is treated as a unit arrow. If the arrows point in many directions, they cancel and r is small. If they point together, they reinforce and r approaches 1.

As K increases, a typical population passes through three qualitative stages:

- I. **Drift:** phases move mainly according to their own natural frequencies.
- II. **Partial Locking:** a group with similar frequencies begins to move together.
- III. **Strong coherence:** most phase differences remain bounded and the flashes form a stable rhythm.

5 A Short Simulation

Euler’s method can approximate the phase evolution. The following compact algorithm repeats the interaction and update steps.

Algorithm 1 Simulate the Kuramoto model

Require: Phases θ_i , frequencies ω_i , coupling K , step size h , and number of steps M

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1: for  $m \leftarrow 1$  to  $M$  do
2:   for each firefly  $i$  do
3:      $q_i \leftarrow \sum_{j=1}^N \sin(\theta_j - \theta_i), \quad v_i \leftarrow \omega_i + \frac{K}{N} q_i$ 
4:   end for
5:   Update all phases simultaneously:  $\theta_i \leftarrow (\theta_i + h v_i) \bmod 2\pi$ 
6: end for
7: return  $\theta_1, \dots, \theta_N$ 
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Each firefly compares its phase with all N phases. Therefore, for M time steps,

$$T(M, N) = \Theta(MN^2) \tag{11}$$

6 What the Model Teaches

The Kuramoto model is deliberately small: it keeps only phase, natural frequency, and coupling. Yet it explains several general lessons.

- ◊ A shared rhythm can emerge without a leader or global clock.
- ◊ The same mathematical language can describe biological rhythms, power-grid generators, chemical oscillations, and distributed timing systems.

The model also has limitations. Real fireflies interact through visible pulses, have restricted fields of view, move through space, and may respond differently to males and females. Those details require richer pulse-coupled or spatial models. A field study and video of natural synchronous swarms are available through the [Science Advances study by Sarfati, Hayes, and Peleg](#).

References

- [1] Renato E. Mirollo and Steven H. Strogatz. “Synchronization of Pulse-Coupled Biological Oscillators”. In: *SIAM Journal on Applied Mathematics* 50.6 (1990), pp. 1645–1662. DOI: [10.1137/0150098](#).
- [2] Raphaël Sarfati, Julie C. Hayes, and Orit Peleg. “Self-Organization in Natural Swarms of *Photinus carolinus* Synchronous Fireflies”. In: *Science Advances* 7.28 (2021), eabg9259. DOI: [10.1126/sciadv.abg9259](#).